A cross-protocol dataset of blockchain oracle incentives and accountability

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Abstract. Blockchain oracles connect smart contracts to external information, but their records use different units and accounting conventions. We present a data resource covering eight oracle branches, with five detailed panels and three narrower extensions. Native observations are linked to protocol rules and reconstructed lifecycles, distinguishing configured incentives, accruals, entitlements, payments and applied losses. UTC annotations preserve event times, epoch intervals and snapshots separately. Two derived products provide daily measurements within documented coverage and a request-level UMA prediction task with candidate inputs, availability records and protocol-resolution labels. Validation examines native identities, economic classifications, settlement accounting and a cross-chain dispute. The resulting resource supports descriptive research on participation, rewards and accountability while retaining differences in mechanism, asset and observation unit. Missing evidence remains explicit, and protocol adjudication is distinguished from independent verification of external facts.

1 Introduction

Blockchain oracles supply smart contracts with information that the ledger cannot observe. Prices and event outcomes enter contract execution through reporting, verification and dispute resolution [1, 2]. Participants may earn rewards, post capital, challenge reports or incur penalties. These arrangements determine both the information a contract accepts and the incentives for providing it [3, 4].

Public transactions do not make these incentives directly measurable. A transfer may combine returned principal with a reward; a negative accounting adjustment may precede any reduction in stake. Disputed requests can traverse several contracts or chains, whose versions, identifiers and accounting rules differ. Treating gross payouts as profit or accruals as realized losses changes the economic quantity being measured. Comparison requires lifecycle reconstruction under the applicable protocol rules.

Oracle mechanisms expose different aspects of economic accountability: proposals and adjudication, staking operations, reward epochs, signed updates or withdrawals. Oracle4SD Atlas selects branches for this variation and the historical evidence available to reconstruct it. The sample supports comparisons of specified activities; it is neither a census of oracle services nor a probability sample of the ecosystem.

The resource covers eight branches: five detailed panels for UMA, Chainlink, Tellor, Flare FTSOv2 and Pyth, and three narrower extensions for Chronicle, RedStone Push and DIA. The common accountability representation contains 105,588,120 typed observations, complemented by 22,723,389 economic components and 19,275,854 records meeting protocol-specific realization criteria. These representations overlap and use different observation units, so their counts cannot be added. Native records accompany the common fields, linking comparative measures to their supporting evidence.

The contribution is a measurement framework that connects protocol rules and native records to observable economic states. By distinguishing configured incentives, accrued adjustments, claimable entitlements, realized payments and applied losses, the resource supports descriptive research on participation, incentives and accountability within documented coverage boundaries. Figure 1 situates the sampled implementations within the wider oracle ecosystem, while Table 3 summarizes the scope of each branch. Comparisons remain conditional on observation units, historical rules and evidence availability. Protocol acceptance and financial execution are therefore interpreted separately from independent verification of external facts.

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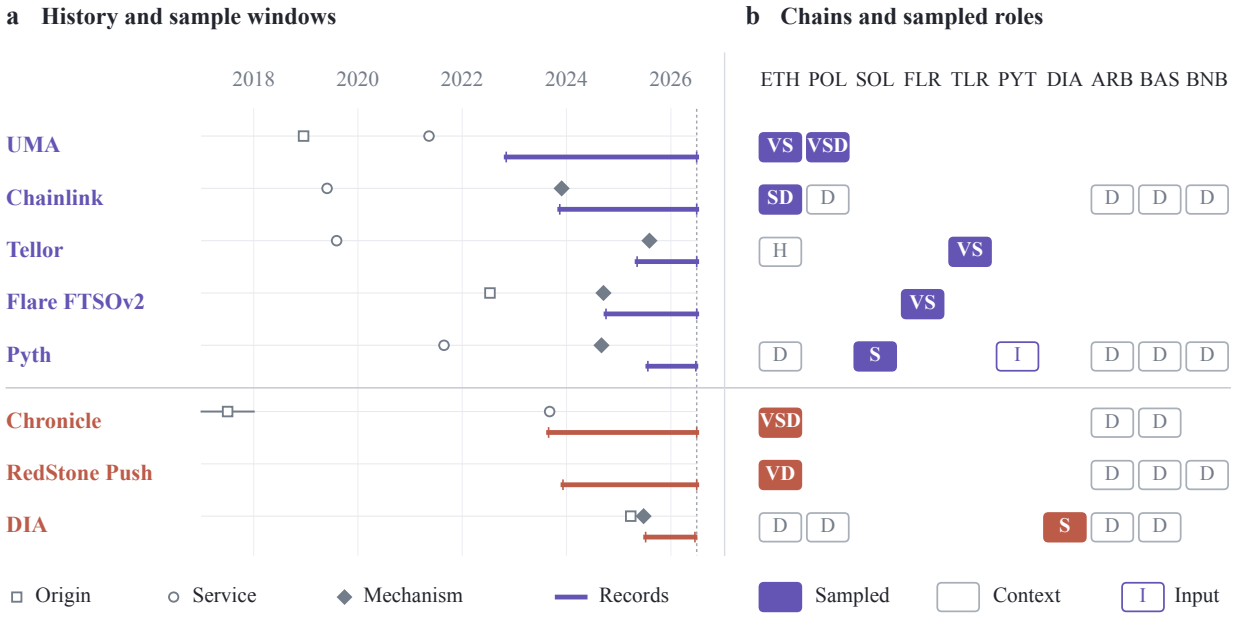


Figure 1. Historical context and sampled scope of Oracle4SD Atlas. Purple denotes five detailed panels; coral denotes three narrower extensions. (a) Symbols mark project or network origins, service openings, and mechanism announcements or access milestones. Coloured segments delimit retained timestamps in one selected table per branch; they do not imply uninterrupted acquisition. The dashed line marks the 30 June 2026 cutoff. (b) Filled cells identify sampled chain roles; grey outlines show selected ecosystem context. V, verification/adjudication; S, economic settlement/accounting; D, delivery; H, historical implementation; I, retained derived inputs. ETH, Ethereum; POL, Polygon; SOL, Solana; FLR, Flare; TLR, tellor-1; PYT, Pythnet; DIA, Lasernet; ARB, Arbitrum; BAS, Base; BNB, BNB Chain. Dates and source evidence are provided in Appendix A.

2 Related work

Research on oracles and research on blockchain datasets approach the same infrastructure from complementary directions. Oracle studies examine how external information is authenticated, aggregated and incentivized; data-resource studies establish reusable representations of transactions, entities and contracts. The present work connects these perspectives by making the relationship between native protocol evidence and economic measurement explicit. Tables 1 and 2 summarize representative contributions without treating their differing units or purposes as directly comparable measures of scale.

2.1 Oracle design, security and economic institutions

Town Crier authenticates external data delivery through trusted hardware; DECO proves properties of data obtained over TLS [1, 5]. The oracle systematization of knowledge reviews trust assumptions, attacks and mitigations [2]. Adaptive conformal consensus provides conditional guarantees for aggregating reports, while SMARTCAT detects contracts associated with price-manipulation attacks [6, 7]. These studies address information security; we reconstruct participant incentives and economic outcomes.

Empirical oracle studies provide a second foundation. A first look at DeFi oracles measures deployed price-oracle behavior, while Demystifying Pythia examines Chainlink usage, pricing and service quality on Ethereum [8, 9]. Economic research complements these observations by analyzing incentives, staking, decentralization and the integration of oracle networks with financial markets [4, 10]. Together, these studies motivate an empirical representation that follows the economic process across participation, challenge, adjudication and settlement. The framework developed here retains those stages where the source evidence supports them, rather than assuming that every mechanism exposes an equivalent lifecycle.

The mechanisms represented in the Atlas expose different empirical objects. UMA and Tellor provide settings for studying contestable reporting, challenge participation and adjudicated outcomes [22, 24]. Chainlink's staking and feed records distinguish economic participation from observable service updates [23],

while Flare FTSOv2 separates reward entitlements, provider conditions and subsequent claims [25]. Pyth's OIS records connect publisher-related incentive states with staking operations and economic transfers within the retained historical window [26]. The narrower Chronicle, RedStone Push and DIA extensions add observations of challenge mechanisms, signed update delivery and staking withdrawals, respectively [27–29]. These branches contribute different parts of the measurement problem; their native records are not interchangeable observations of a universal oracle process.

Oracle economics and empirical studies motivate questions about incentives, monitoring, participation and service provision [4, 8–10]. The Atlas connects these questions to observation units, historical rules and economic stages. Documentation describes intended mechanisms; execution records show observed behavior. Signatures, consensus outcomes and payments do not independently establish external truth, and links to mechanism design do not establish theoretical equilibrium or security guarantees.

Table 1. Oracle research and the measurement problem addressed here. "Conceptual" denotes a mechanism or taxonomy rather than a deposited cross-protocol economic panel. Contributions are compared as stated.

Study	Evidence focus	Representation	Economic coverage
Town Crier [1]	Authenticated delivery	Trusted-hardware oracle architecture	External-data delivery; no cross-protocol economic lifecycle panel.
DECO [5]	Web-data provenance	Cryptographic proofs for TLS data	Data provenance and privacy; no reconstruction of oracle compensation or losses.
Oracle SoK [2]	Security and trust	Taxonomy of designs, attacks and assumptions	Conceptual treatment of economic security and manipulation.
DeFi oracle measurement [8]	Deployed price oracles	Reported values, deviations and operation	Selected operational measurements rather than harmonized incentive states.
Demystifying Pythia [9]	Chainlink on Ethereum	Usage, pricing and service quality	System-specific economic and usage measurements.
Oracle economics primer [4]	Incentives and institutions	Economic framework for oracle mechanisms	Conceptual analysis of staking, decentralization and tokenomics.
This resource	Participation and accountability	Native lifecycles with typed economic components	Five detailed panels and three extensions; realized economics only where observable.

2.2 Reusable blockchain data resources

Elliptic represents Bitcoin transactions for illicit-activity classification, while Elliptic2 labels subgraphs of entity clusters [11, 12]. Chartalist provides datasets for UTXO- and account-based blockchains; temporal Bitcoin datasets support entity and transaction graph analysis [13–15]. EX-Graph connects Ethereum and X, and Graphs of Graphs preserves relationships within and across token networks [16, 17]. Oracle research also requires the economic meaning of each request, report, entitlement and balance change.

Other resources link NFT and memecoin markets to network and visual features [18, 19] or organize fine-grained DeFi events for forecasting [20]. DIVE separates pre- and post-deployment contract representations for vulnerability analysis [21]. We likewise retain multiple representations of an object, but interpret oracle participation and financial accountability through protocol-specific accounting rules.

Comparative oracle research needs shared concepts tied to native roles, timing and economic states. Our layered representations distinguish announced incentives from claimable entitlements, transfers from participant gain, and institutional decisions from external truth. Validation preserves these distinctions.

Table 2. Representative blockchain datasets and analytical units. Counts refer to different objects and cannot be compared as sample sizes. Oracle4SD Atlas complements these resources with protocol-specific economic stages.

Resource	Observation unit	Published scope	Analytical purpose
Elliptic [11]	Bitcoin transaction	203,769 nodes; 234,355 directed edges	Illicit-transaction classification and financial forensics.
Elliptic2 [12]	Entity cluster and labeled subgraph	49.3 million clusters; 196.2 million edges; 121,810 labeled subgraphs	Money-laundering patterns at subgraph level.
Chartalist [13]	Dataset-specific graph, address or transaction	Multiple UTXO- and account-based blockchain datasets	Reusable data access, graph construction and machine-learning benchmarks.
DIVE [21]	Ethereum smart contract	22,330 contracts; pre- and post-deployment representations	Vulnerability analysis with lifecycle-specific features.
This resource	Typed protocol-native observation	105,588,120 harmonized observations; eight branches	Incentive and accountability measurement, with native units and economic realization states retained.

3 Methods

Construction links protocol evidence to economic interpretation before defining common analytical fields. Figure 2 summarizes source registration, native decoding, lifecycle reconstruction and harmonization. Unresolved identities, open lifecycles and unobservable outcomes remain explicit. The common schema supports comparison while preserving protocol-specific meanings.

3.1 Scope, selection and historical boundaries

The discovery registry contains 56 oracle or related mechanism categories and 366 protocol–oracle relationships. It supports source discovery and ecosystem description, not probability sampling. Eight branches are selected for their economic mechanisms, identifiable historical records and scope for reconstruction. Each branch denotes a specified mechanism and implementation, including the chain roles observed in its records.

UMA, Chainlink, Tellor, Flare FTSOv2 and Pyth provide five detailed native panels. Chronicle, RedStone Push and DIA add narrower observations of challenge, update delivery and staking activities. Panel breadth implies neither complete histories nor equivalent variables or protocol quality. Only registered versions and services are included. Table 3 summarizes branch scope; Appendix A lists sources and exclusions.

Chain roles are distinguished by function. A verification or adjudication chain records the relevant acceptance or dispute-resolution process; an economic settlement chain records the associated payment or applied balance change; and a delivery chain records provision of the oracle output to a consuming application. A single network may serve several roles. In the linked UMA disputes studied here, Polygon hosts the OOv2 lifecycle and observed local settlement flows, whereas Ethereum hosts VotingV2 adjudication. Linking these records establishes a relationship between protocol objects and does not, by itself, establish a cross-chain token transfer. The eight branches consequently do not denote eight independent blockchains.

The common upper observation cutoff is 30 June 2026 at 23:59:59 UTC. Table-level temporal support is documented separately from protocol history in Appendix B. The temporal register identifies deployment-bound scans, implementation changes, retained-history limits and cohort restrictions without treating a first event as a deployment date. Lower boundaries vary with deployment, implementation changes, historical source availability and collection limits. Project founding dates, production-service launches, deployment boundaries, scanned intervals and first observed events are distinct quantities. An interval between the first and last retained events does not establish uninterrupted observation. Transaction histories and retained state panels may also have different temporal support, as in the Pyth OIS branch. Historical interpretation requires an implementation applicable to the observation; unresolved source bindings and acquisition gaps remain limitations. Current protocol documentation supplies architectural context [22–29], while Appendix A

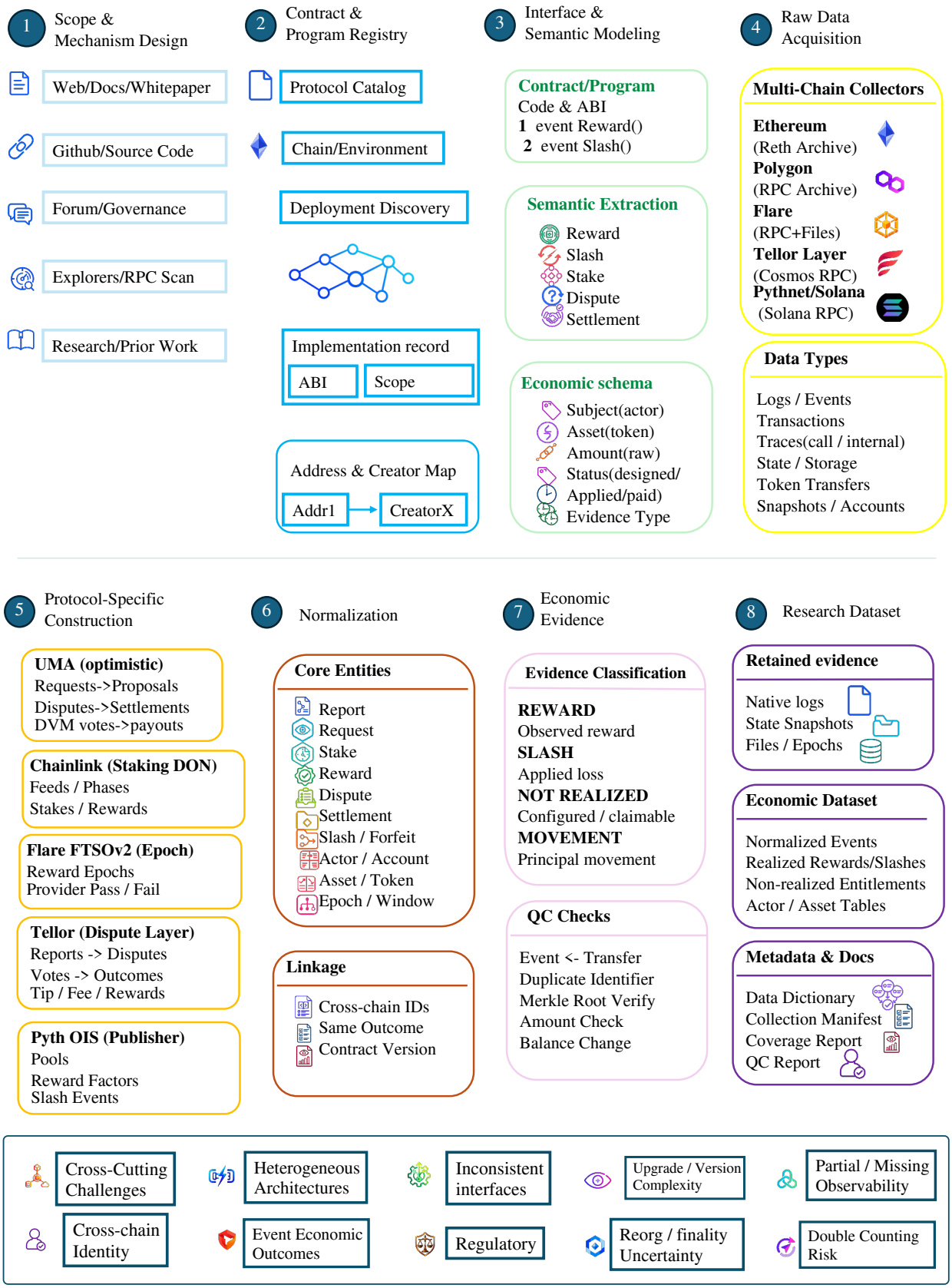


Figure 2. Construction and validation workflow. Stages 1–4 register sources, define native semantics and acquire evidence; stages 5–8 reconstruct supported lifecycles and economic states for analysis and documentation. The lower strip lists measurement challenges. Checks require evidence; protocol decisions do not establish external truth.

distinguishes that context from retained historical evidence.

Table 3. Measurement roles, implementation scope and historical boundaries of the eight oracle branches. Coverage refers to registered sources, not all services of a protocol. The upper cutoff is 30 June 2026, 23:59:59 UTC. Appendix A separates deployment, scanning and retained-event boundaries.

Branch	Implementation and chain	Registered coverage	Measurement role and main exclusion
UMA	Ethereum VotingV2; Polygon OOv2 and adapters	DVM blocks 16,697,232–25,433,938; Polygon contract-specific starts through 89,439,336	Proposals, disputes, adjudication and settlement. Other UMA versions and deployments are outside the registered scope.
Chainlink	Ethereum Staking v0.2; ETH/USD proxy and feed phases	Staking from block 18,572,190; registered sources through 25,433,938	Staking economics and selected feed updates. Other feeds, networks and services are excluded.
Tellor	`tellor-1` modules, ABCI events and block results	Heights 1–19,890,860; first report at height 317	Reports, disputes, votes, payments and restrictions. Legacy Ethereum Tellor is excluded.
Flare FTSOv2	Flare RewardManager, VoterRegistry and reward artifacts	Closed reward epochs 228–410; cutoff block 64,054,789	Entitlements, provider conditions and claims. Table coverage differs; aggregate claims may resist attribution.
Pyth	Solana OIS transactions and retained Pythnet-derived inputs	Transaction archive from 11 September 2024; 49 retained closed state epochs	Publisher–epoch states, staking and transfers. State coverage is shorter; other products and delivery networks are excluded.
Chronicle	Registered Ethereum validator and challenge contracts	Registered log scan 0–25,433,938; deployment dates are separate	Reports, challenges and challenge rewards. Rewards are not a complete publisher-compensation ledger.
RedStone Push	Registered Ethereum Push adapters and feeds	Registered log scan 0–25,433,938; deployment dates are separate	Signed publication and Push updates. Other delivery modes and full AVS economics are excluded.
DIA	Registered Lasernet staking contracts	Deployment block 3,874,186 through cutoff 31,302,665	Withdrawals, returned principal and rewards. This is not a complete price-production or delivery audit.

3.2 Source acquisition and native decoding

For each branch, a source register identifies the network, contract or program, implementation, collection interval and retrieval interface. EVM-compatible branches use historical logs, supplemented where necessary by transactions, receipts, block metadata and token-transfer evidence. ERC-20 transfers are retained when they help separate principal, fees and rewards from a settlement event. Tellor instead combines tellor-1 module state, ABCI events and block results; multiple reports within one block produce a one-to-many expansion into 82,866,640 report observations. Pyth evidence is located by Solana transaction signatures and outer or inner instruction positions. Flare combines on-chain records with official reward-calculation artifacts, preserving the distinction between entitlement and claim.

Decoding follows the registered version and cutoff. Polygon UMA has 626,578 raw records, including five unknown event topics retained only in the raw collection. Chainlink staking has 175,015 collected logs, yielding 174,890 decoded events and 125 unclassified records. Unclassified records do not indicate absent economic behavior. Tables retain native keys, integer amounts, asset identifiers and decimal precision.

3.3 Lifecycle reconstruction and cross-chain linkage

Decoded records are joined where the native mechanism identifies a relationship between successive states. UMA requests link proposals, disputes, DVM adjudication and settlement through reconstructed request

identifiers. For requester r, identifier i, timestamp t and ancillary data a, the identifier is

$$q = \text{Keccak256}(\text{pack}(r,i,t,a)), \tag{1}$$

where pack denotes the protocol's packed ABI encoding. The reconstructed OOV2 panel contains 42,676 request rounds: 41,826 settle by the cutoff and 850 remain open. Other mechanisms require different objects. For example, 46,723 retained Chainlink ETH/USD answer updates define 46,722 adjacent service windows, and 13 Tellor jail events are paired with 13 release events to reconstruct reporter restrictions.

Cross-chain identity is established through version-aware ancillary-data transformations and native request construction. A UMA link receives Grade A only when the registered bridge or adapter rule identifies one unique DVM request. Of 835 attempted links, 832 meet this deterministic definition and three remain unmatched. The grade describes the identity of a reconstructed object; it does not certify the factual correctness of the oracle's decision. Retaining unmatched and unfinished objects prevents the apparent completeness of an analytical table from replacing evidence about the underlying process. The worked case links request, proposal, dispute, DVM resolution and Polygon settlement. Each endpoint is located by its chain, transaction and log position. The retained evidence supports the object linkage and elapsed intervals; intermediate relay operations are represented separately where their transactions are not retained.

3.4 Economic states and settlement accounting

An economic observation is classified by its state and evidentiary basis. Table 4 distinguishes configured incentives, accrued rewards or penalties, claimable entitlements, realized payments, applied losses, principal, refunds and fees. The distinctions are protocol-specific. UMA challenge status requires native challenge evidence, and settlement is not inferred merely from a transfer-like event. Tellor SUPPORT outcomes refer to acceptance of the challenge and rejection of the challenged report. Negative DVM accruals remain distinct from an observed reduction in staked principal. Pyth reward rates are not token reward amounts, and Flare participation restrictions are not automatically monetary losses.

UMA settlement provides a concrete accounting example. Let B be the effective bond, F the final fee and R the configured reward, all measured in the smallest integer unit of the same asset. Posted principal is $P = B + F$. For disputed requests, the identifiable winner component is $W = B - \lfloor B/2 \rfloor$, while the protocol component is $F_{\text{protocol}} = \lfloor B/2 \rfloor + F$. Expected gross payouts follow the registered contract and adapter rules:

$$\begin{aligned} G_{\text{undisputed}} &= B + F + R, \\ G_{\text{disputed}} &= B + F + W + R_{\text{settlement}}. \end{aligned} \tag{2}$$

Here $R_{\text{settlement}}$ is the reward attributable to settlement under the applicable adapter; refunds and rollovers are not automatically participant income. Exact integer reconciliation requires $\Delta_{\text{payout}} = G_{\text{actual}} - G_{\text{expected}} = 0$.

In Section 6.6, a 1,000-unit payout comprises 750 in principal and 250 in gain before costs. A configured five-unit reward is refunded or rolled over as observed. Net profit requires cost data. UMA settlements are checked against token movements, Chainlink separates transfer-requiring events from accounting adjustments, and Flare reconciliation retains only components supported by the claim evidence.

3.5 Harmonization, missingness and documentation

Common fields retain protocol, unit, role, actor or beneficiary, lifecycle identity, time, asset, economic stage, source and evidence status. The accountability layer contains 105,588,120 typed observations; the economic layer contains 22,723,389 components, including 19,275,854 meeting branch-specific payment or applied-state criteria. These overlapping layers describe the same underlying processes at different levels.

Timestamp availability differs from coverage: a valid clock on every retained row does not prove complete collection, and a missing clock does not imply an absent event. Epoch and snapshot clocks remain distinct from canonical event time. Missing values can indicate inapplicability, unexposed quantities, incomplete collection, decoding problems or unfinished lifecycles. They are not automatically zero. Where supported, schemas distinguish complete, partial, verified-zero, unknown, unavailable and not-applicable coverage; right censoring is recorded separately. Deduplication uses native keys and preserves legitimate multiplicity across representations and economic components. Table 5 summarizes these operations.

Table 4. Economic states distinguished in reconstruction. Each classification requires the evidence specified by the originating protocol. A lifecycle need not expose every state.

State	Economic meaning	Interpretation rule
Configured incentive	Reward specified in protocol parameters	A design parameter; not automatically participant income.
Accrued reward or penalty	Calculated entitlement or liability	Distinct from subsequent payment or application; counting both may duplicate one change.
Claimable entitlement	Amount eligible for a claim	An unclaimed entitlement is not a zero payment or a realized transfer.
Paid reward	Observed transfer or realized protocol credit	Realized only when qualifying cash-flow or state evidence is present.
Applied slash	Reduction applied to a participant's balance or principal	Requires observed application, beyond a penalty calculation or accrual.
Principal deposit or return	Capital posted and subsequently recovered	Kept separate from rewards and fees; returned capital is not profit.
Refund or rollover	Incentive returned or carried to another state	Not attributed to the current participant without supporting evidence.
Protocol fee	Amount charged or retained under protocol rules	Separated from participant reward and principal.

Table 5. Processing operations and their outputs. Rows describe transformations rather than statistically independent populations: operations may expand, aggregate, join or decompose their inputs.

Operation	Input and rule	Output
Source registration	Network, contract or program, historical version, interval and interface	Branch-specific collection specification.
Acquisition and qualification	Logs, transactions, receipts, block or module state, signatures and protocol artifacts; cutoff and decoding rules	Qualified source records, with unresolved decoding outcomes retained.
Native decoding	Protocol event and state definitions; integer amounts and native keys	Protocol-native tables.
Lifecycle reconstruction	Request, dispute, claim, epoch or other native object relationships	Supported lifecycle objects and open cases.
Cross-chain linkage	Version-specific identifier and ancillary-data transformations	Deterministic matches, link grades and unmatched cases.
Economic-state mapping	Native lifecycle and accounting evidence	Configured, accrued, claimable, paid, applied, returned, refunded and fee components.
Funds-flow reconciliation	Expected quantities against observed transfers or applied state changes	Realization flags and accounting differences.
Harmonization	Shared concepts with native units and source links retained	105,588,120 typed accountability observations.
Realization filtering	Protocol-specific evidence conditions applied to 22,723,389 economic components	19,275,854 qualifying realized components.
Documentation	Dictionaries, schemas, coverage information, source registers and validation results	Interpretable data and reconstruction materials.

Documentation records native identifiers, source registers, input inventories, schemas, variable definitions and validation rules. Quality controls assess selected key relationships, lifecycle orderings, cutoff conditions, economic identities and deterministic links. The accompanying artifacts identify the tested populations and the evidence available for reconstruction. Their traceability supports inspection of specific mappings, while leaving incomplete historical acquisition and unresolved source ancestry visible.

3.6 Temporal harmonization

Native timestamps are expressed in UTC while retaining their original values and precision. Each time field is documented by its source, unit and role in the protocol. The field `event_time_utc` identifies when an event was recorded or executed. Request-reference timestamps, oracle payload times and scheduled deadlines are represented separately because they can refer to a different point in the lifecycle. Records sharing a timestamp retain their native block, transaction, log or instruction order.

Unix-second and millisecond values are converted without rounding. Textual timestamps are interpreted using their recorded timezone; timezone-naive values are treated as UTC where this convention is established by the source. Missing EVM event times are recovered only when another retained record identifies the timestamp of the same block on the same chain. Conflicting matches are rejected and unmatched observations remain undated. The provenance of recovered timestamps is recorded with the data.

Epoch observations retain their start and end times as half-open intervals $[a_e, b_e)$. Payment, snapshot and retrieval times describe separate events and are stored independently. In particular, an epoch-end label does not establish when a reward became claimable or was paid. Historical state recovered from a later snapshot also requires care when assessing information available at a past decision time.

Daily windows begin at midnight UTC, and weekly windows run from Monday to the following Monday. Epoch totals remain attached to their native support intervals. Daily aggregation therefore requires an event-level measure or an explicit allocation rule, together with evidence that the relevant period was observed. Appendix B describes the native clocks and temporal limitations of each branch.

3.7 UMA prediction-task construction

The prediction task asks whether UMA's eventual resolution differs from the proposed price. Each observation is one request, and the decision time T_0 is the proposal event. For proposed price p_i and resolved price z_i, the label is $Y_i = \mathbf{1}\{z_i \neq p_i\}$. Native values are compared as integers. Indeterminate and early-expiration values retain their own classes; neither is recoded as a binary answer.

Eligibility requires a challenged request, a unique deterministic DVM link, consistent resolution values and a reconciled settlement transfer. Because selection is retrospective, the task concerns adjudication among selected disputes, not challenge risk for new requests. Challenge, resolution and settlement fields supply outcome evidence and are excluded from predictive inputs.

The sixteen candidate features describe incentives, timing, ancillary-data length, participant history and adapter identity. Proposer and requester histories are calculated separately from requests settled before the current proposal in native execution order. Rates are missing without prior settlements. Proposal latency begins at the request event; liveness ends at expiry. Parameter-derived features require evidence of availability at T_0 and carry a separate status. Appendix F defines the features and their information boundary.

3.8 Daily economic measurements

Daily observations use midnight-to-midnight UTC intervals and retain branch, implementation, actor role, asset, unit and economic stage. Feasibility distinguishes derivable measures, additional acquisition requirements, inapplicable quantities and unresolved cases. Appendix D defines aggregation and denominators.

Payments enter the daily product only under the relevant realization rule. UMA rewards are separated from returned principal and forfeited bonds. VotingV2 net stake applications retain their own economic stage and are not added to request-level accruals; Tellor withdrawals into stake and Pyth transfers to stake custody retain those destinations. Flare fee burns are excluded from participant payments. Chainlink claims require a matched transfer and an execution timestamp. An unobserved day is left unfilled, and totals over retained records are not presented as complete protocol totals.

Participation measures count addresses within a named role. Reward concentration is calculated among recipients with positive observed payments, separately by asset and economic stage. For completed UMA requests, delay runs from the request event to settlement, and escrow exposure integrates observed balances between successive transfers. Only unambiguous closed request flows with nonnegative intermediate balances are included; the measure includes funded rewards while held and does not represent unresolved capital obligations. Flare signing weights and thresholds are retained on their native epoch intervals; they are not interpreted as daily payments or token collateral.

4 Data records

Ten logical products cover eight protocol branches, the UMA prediction task and daily measurements. Each may contain several linked tables. Common representations support comparison; native tables preserve mechanism-specific timing and accounting. Table 6 lists the principal components, with types, definitions and relationships documented in the catalogue and dictionaries.

Table 6. Data organization. Tables may contribute to several representations; row counts are not additive.

Component	Form	Contents
Eight native branches	Parquet and retained source artifacts	Requests, reports, votes, claims, transfers, policies and state observations.
Common representations	Typed Parquet projections	Shared accountability fields and economic components with native source references.
Temporal annotations	UTC fields and dictionaries	Event times, reference clocks, epochs, snapshots and missing-time status.
UMA prediction task	Request and availability tables	Protocol-resolution label, sixteen candidate features and retrospective selection rules.
Daily measurements	Daily and epoch companion tables	Typed counts and amounts, recipient aggregates, completed-request delays, capital exposure and policy weights.
Supporting evidence	Ledger and case tables	Old/new semantic values, source locators, case intervals and cashflows.
Metadata and reproduction	Catalogues, dictionaries and code	Ten logical products, physical recordsets, fingerprints and executable examples.

4.1 Native records and table relationships

Native tables preserve the object meaningful to the originating protocol. The UMA OOV2 panel contains 42,676 request lifecycles, while the Tellor micro-report table contains 82,866,640 reports. Other examples include 18,468,735 Flare claim events and 518,560 decoded Pyth OIS instructions. These quantities count requests, reports, claims and instructions, respectively. A comparison that treats them as equivalent oracle events would confuse differences in data granularity with differences in institutional activity. Within a branch, related evidence can also be stored separately: UMA request-level lifecycle and payout fields link to token-flow records rather than repeating each transfer inside the request table.

Identifiers follow the strongest available native structure. EVM logs use chain, transaction and log position; Tellor relies on module-specific report metadata; Solana evidence uses signatures and instruction positions. Table 7 summarizes these keys and their relationships. Checks on the named UMA, Chainlink and cross-chain linkage tables assess selected primary-key and foreign-key constraints. A one-to-many relationship can represent several events within a lifecycle or several components within one settlement. It is therefore a relationship to interpret, rather than a duplication to remove automatically.

4.2 Common accountability and economic representations

The common accountability table contains 105,588,120 rows and 50 columns. Each row projects a native observation while retaining protocol, unit, role, source and evidence status. Requests, reports, disputes,

Table 7. Native units, identifier construction and logical relationships. One-to-many relationships describe scientific objects, not mandatory duplication in every physical table. Native ordering is authoritative; physical row order is not.

Unit	Identifier	Relationship and interpretation
EVM log	Chain, transaction hash and log index	A transaction can contain several logs. Order by block and log position; retain receipt and block evidence.
UMA OOV2 request	`oo_request_id`; requester, identifier, timestamp and ancillary data	One request links several lifecycle events and transfers. Request, proposal, dispute and settlement times remain distinct.
UMA DVM request or voter record	`dvm_request_id`; voter and source-event location for actor-level records	One request can link many voters and adjustments. Accrual and applied stake change are different observations.
Tellor report	`query_id`, reporter and `meta_id`; block-position namespace for the common key	Multiple reports can contribute to an aggregate. Native module metadata replaces a universal EVM transaction/log key.
Flare entitlement or claim	Epoch, beneficiary and claim type; transaction/log position for a claim	Entitlements and claims are not universally one-to-one. Epoch end differs from claim time.
Pyth instruction or transfer	Signature and `outer_instruction_index`; inner position for token transfers	One transaction can contain several instructions and transfers. Reconstruct time from block time and instruction positions.
Common observation	`accountability_event_id`, protocol namespace, `native_table` and `accountability_unit_id`	Native-to-common mapping can be one-to-many. The retained native unit defines the statistical object.
Economic component	`evidence_id` and originating table/event identity	One settlement can yield several components. `do_not_sum_group` identifies related components at risk of double counting.

epochs, staking observations and service windows remain distinct units. Cross-protocol aggregation must therefore retain protocol, native table, unit type and unit identity to establish comparability.

Economic components are represented in a separate 22,723,389-row, 28-column table. Its fields record the mechanism, stage, actor, counterparty, amount, asset, source object, evidence class and validation status. A stricter 19,275,854-row representation uses the same physical schema but requires qualifying evidence of payment or an applied economic change. It overlaps the broader component table and is not an independent sample. Stage, actor, asset and the documented grouping of related components must be considered together: adding an accrued penalty to a later applied slash can count the same economic process twice.

4.3 Dictionaries, types and interpretive states

Documentation operates at two levels. The conceptual dictionary defines 33 economic variables: 28 core measures and five supporting variables. The core measures comprise four participation or concentration variables, nine incentive or friction variables, and fifteen outcome or accountability variables. Supporting entries identify the episode, decision time, challenge deadline, cross-chain link grade and provenance. Their definitions specify units, derivation, decision-time availability, evidence class, missingness and validation rules. The physical dictionary describes the columns in each table, including the derived products and their supporting evidence. Conceptual variables and physical columns serve different purposes; the catalogue records their relationships rather than treating their counts as alternative estimates of dataset dimensionality.

Table 8 summarizes selected field families. Raw monetary amounts are stored as integer-valued strings and interpreted with their asset and decimal fields, avoiding loss from floating-point conversion. The common representation includes integer Unix-second labels. Separate UTC annotations preserve native timestamp precision and distinguish event times from epoch boundaries and snapshots. JSON schemas define required fields and controlled values, including the 85-field cross-chain observation schema. Where supported, coverage distinguishes complete, partial, verified-zero, unknown, unavailable and not-applicable conditions; right censoring separately identifies unfinished lifecycles. Schema checks establish consistency with these

tested definitions, rather than proving every historical interpretation correct.

Table 8. Selected field families. Dictionaries define the schema; amounts require asset and decimal fields.

Field family	Representative fields	Type and analytical meaning
Identity and mechanism	`oracle_network`, `mechanism_family`, `native_table`, `security_chain`, `delivery_chain`	Strings identifying the mechanism, native table and chain context.
Observation unit	`accountability_unit_type`, `accountability_unit_id`, `event_granularity`	Strings or controlled values defining what a row represents.
Time	`event_time_unix`, `reward_accrual_time_unix`, `reward_payment_time_unix`	Integer-second labels in the common representation; separate UTC annotations distinguish event, epoch, payment and snapshot times (Appendix B).
Actors and roles	`actor`, `actor_role`, `counterparty`, `counterparty_role`	Native addresses or identifiers and interpreted roles; not natural-person identities.
Reward	`reward_class`, `reward_amount_raw`, `reward_asset`, `reward_asset_decimals`	Integer string, asset identity and decimal precision. Economic stage determines interpretation.
Penalty and forfeiture	`penalty_accrued_raw`, `principal_slashed_raw`, `bond_forfeited_raw`, `fee_forfeited_raw`, `reward_forfeited_raw`	Integer strings with asset metadata; distinguishes accrual, applied principal loss and other forfeitures.
Principal	`principal_locked_raw`, `principal_returned_raw`, `principal_asset`	Integer strings and asset metadata; separates exposed or recovered capital from gain.
Outcome basis	`truth_basis`, `outcome_status`, `external_truth_available`	Strings and booleans describing the basis for an outcome; protocol resolution is not external ground truth.
Source and rule	`source_event`, `source_tx`, `source_block`, `source_contract`, `rule_id`, `parameter_version`	Strings and integers linking observations to native evidence and the interpretation rule.
Realization	`economic_kind`, `economic_evidence_class`, `realization_status`, `amount_raw`, `signed_amount_raw`, `cashflow_verified`, `state_delta_verified`	Controlled values, integer strings and booleans recording the economic stage and evidence for realization.

Curated tables use Zstandard-compressed Apache Parquet; metadata use UTF-8 CSV or JSON, and source packages use JSONL or compressed JSONL. The revised catalogue distinguishes the earlier deposit's 95 tables and application artifacts from the new task, daily measurements and evidence ledgers. Registers, inventories and validation identify acquisition evidence, reconstruction inputs and outputs.

4.4 Temporal metadata

Native tables retain the original timestamps and identifiers alongside UTC fields for the clocks supported by each source. The common accountability and economic representations use companion tables joined by their retained record identifiers. These companions distinguish execution times, epoch boundaries and missing clocks without altering the original observations. Their definitions specify temporal meaning, precision and the reason a time is unavailable. Analyses using the common representations should use these supported UTC clocks; the earlier integer-second labels remain for provenance.

Temporal metadata record each table's available interval and missing timestamps. Execution times, epoch

intervals, deadlines and snapshots retain distinct meanings, allowing comparable periods to be selected without treating every timestamp as an instantaneous event.

4.5 Derived products and supporting evidence

The UMA task contains one row per eligible request, a sixteen-field feature dictionary and a feature-level availability record. The daily product contains typed measurements, recipient-level payment aggregates, completion delays and observed escrow exposure, together with a separate historical Flare signing-policy table. The latter retains epoch support because signing weights are defined by the policy interval.

The correction ledger records old and revised field values for identifiable native objects. The cross-chain case retains the event register, elapsed intervals, cashflow components and unresolved evidence links. These supporting tables document how the measurements were obtained; their rows are not additional independent oracle observations. Appendix C describes the ten products and their relationships.

5 Data overview

Figure 3 illustrates the temporal support and units of selected measurements. Daily participation and payments are available for different retained populations, while UMA delay describes completed requests and Flare concentration describes epoch-specific signing policies. These quantities are interpreted within their branch, role and observation window. Open requests remain in the lifecycle records, days without eligible retained events remain unfilled, and policy weights are kept separate from token amounts. The broader feasibility assessment in Appendix D identifies which comparisons require additional evidence.

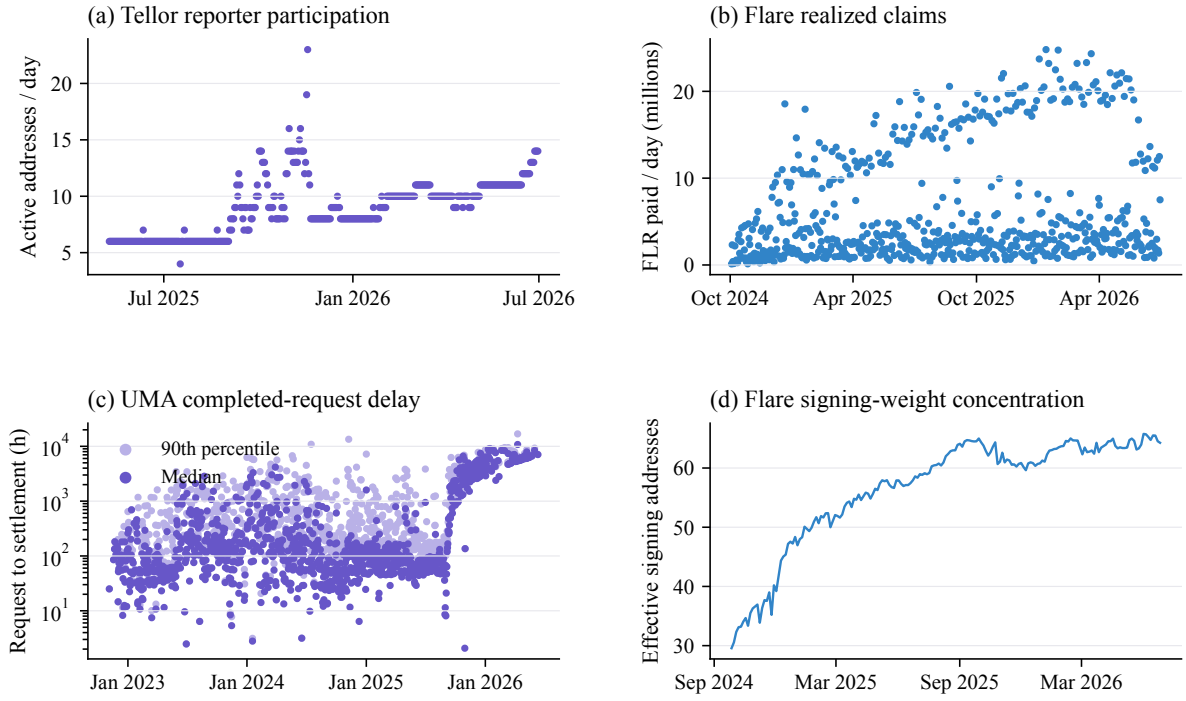


Figure 3. Selected measurements from the daily product and its epoch-level companion. (a) Distinct Tellor reporter addresses with retained reports on each observed day. (b) Flare claims in FLR, excluding fee burns. (c) Median and 90th-percentile request-to-settlement delay among flow-reconciled UMA USDC.e requests completed on that day; the vertical axis is logarithmic. (d) Effective signing-address count, $1/\sum_i w_i^2$, using normalized Flare signing-policy weights within each retained epoch. Address counts do not identify independent entities.

6 Technical validation

Validation tests structural, economic and computational properties within specified populations. Readable files, unique identifiers, balanced settlements and reproducible calculations establish different properties;

none alone proves complete acquisition or external truth. We distinguish documented checks, direct table checks, historical protocol validation and replay from retained inputs.

6.1 Structural consistency and economic semantics

Structural checks examine source identities, native keys, table relationships and lifecycle order. Request-level records and voter-level accounting entries are tested separately because their units differ. Cross-chain checks test whether linked records identify the same protocol object.

Economic classification is checked against the historical implementation, the recorded execution and the financial quantity being measured. For each correction, an object-level record preserves the previous value, the revised value, the source locator and the applied rule. Source binding is strongest for the retained VotingV2 implementation; the OOv2 and Tellor bindings remain qualified, as described in Appendix E.

Table 9 reports classification errors and corrections for UMA challenges and settlements, Tellor outcomes and DVM penalty accruals. Corrected UMA winner-reward components total 201,500 USDC.e and 1,250 USDC, kept separate by asset. These reclassify existing settlements rather than add payments. Of thirteen Tellor disputes, three support the challenge and have positive applied slashes; ten reject it and have no positive applied slash under the recorded checks. SUPPORT denotes a decision for the disputer.

Table 9. Economic-semantic checks and correction evidence. Native populations may overlap across rules.

Rule	Evidence and interpretation	Population	Before / after
UMA challenge	Dispute event and nonzero disputer; a proposal alone is insufficient.	42,676	40,992 / 0
UMA settlement	Settled lifecycle and matched token movement. Open requests remain unsettled.	42,676	850 / 0
UMA winner gain	Contract payout rule and settlement transfer; returned principal is separate.	42,676	834 / 0
Tellor outcome	SUPPORT accepts the challenge and rejects the report.	13	13 / 0
Tellor applied loss	Final dispute outcome; initiation escrow is distinct from applied loss.	13	10 / 0
DVM accrual	Signed request accrual; an applied stake change requires separate evidence.	2,663,684	947,128 / 0

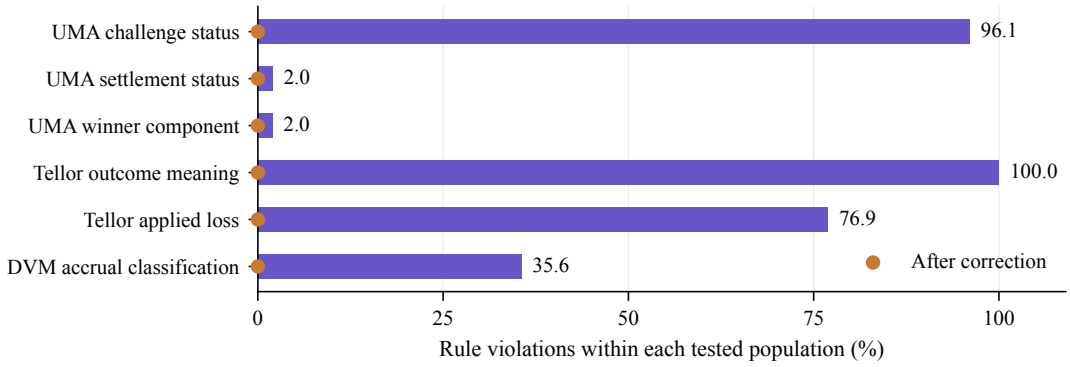


Figure 4. Economic-semantic corrections within six separately defined populations. Bars show the proportion violating each rule before correction; orange markers show the corresponding proportion after applying the rule. Populations and rule evidence are specified in Table 9. A request may be affected by more than one rule.

In the DVM population, 947,128 negative accruals had been classified as applied principal losses without evidence of a corresponding reduction. Reclassifying them as penalty accruals preserves the accounting state without asserting realized loss. Valid records still require interpretation under the mechanism's rules.

6.2 Accounting reconciliation and object identity

All 41,826 settled UMA requests satisfy the recorded integer payout identity,

$$\Delta_{\text{payout}} = G_{\text{actual}} - G_{\text{expected}} = 0, \tag{3}$$

with the associated transfer-matching checks passing under the applicable contract and adapter rules. Chainlink checks identify 51,372 qualified staking and reward records: 50,772 require transfers and match the recorded token movements, while 600 are accounting-only adjustments. The historical Flare report documents Merkle-root and claim reconciliation for 183 closed reward epochs. Those retained results are distinguished from a fresh rerun of the complete chain-level validation.

Of 835 UMA cross-chain link attempts, 832 meet the deterministic Grade-A rule and three remain unmatched. Pyth replay checks reward-state calculations over 49 retained closed epochs. These results support tested links and accounting; completeness, row ancestry and external truth remain unverified.

6.3 Computational reproducibility

A worked UMA case links archived events to a request lifecycle and reconciles economic records with token movements. Repeated executions check consistency for identical inputs and code.

These evaluations concern reconstruction from retained evidence. Reacquiring historical data also depends on archive access, interface behaviour and the original collection boundaries. The acquisition records document those conditions separately from the computational checks.

6.4 Temporal consistency

Temporal checks compare UTC conversions with native values and verify observation order. Millisecond clocks preserve fractional seconds. Native and UTC selections agree for the tested daily and weekly windows. Epoch checks verify retained boundaries and the assignment of observations at interval endpoints.

Missing timestamps are assessed separately from event absence. Exact block matching improves temporal availability for some EVM records, while unresolved observations retain a missing-time status. Appendix B summarizes these limitations and the distinction between timestamp availability and historical coverage.

6.5 Daily aggregation and prediction-task checks

Daily validation checks that grouped event counts and integer monetary totals agree with their eligible native inputs. Escrow exposure is checked by comparing daily balance-duration totals with the integral over the request's observed transfer intervals. A valid timestamp is required for assignment to a day; eligible observations without one remain in the source population and in the missing-time accounting.

The prediction-task checks examine request uniqueness, cohort membership, target construction and feature timing. Role-history counts and rates are compared with a separate direct selection of prior settlements. The checks also preserve the distinction between a native event observed by the proposal and a historical parameter whose availability requires additional source binding. Feature distributions and missingness are provided in Appendix F to support inspection of the task data.

6.6 Worked cross-chain settlement

The worked dispute links four Polygon transactions to an Ethereum VotingV2 resolution. Polygon records the request, proposal, challenge and local settlement. Checks verify receipt status, blocks, event positions and request identity. Figure 5 presents timing and cashflows; Appendix G lists events and relay gaps.

Proposal occurs about 7.7 hours after the request, followed shortly by a challenge. DVM resolution follows about 92.2 hours later, and Polygon settlement follows after roughly 22 minutes. These endpoints cannot separate transport, scheduling and adjudication. Early expiration retains its nonbinary meaning.

Each party posts 750 USDC.e. The winner receives 1,000 USDC.e: 750 in returned principal and 250 in gain. The fee recipient receives 500. The configured reward is refunded or rolled over separately. Flows balance, but net profit requires costs. Cross-chain identity does not establish a bridge transfer.

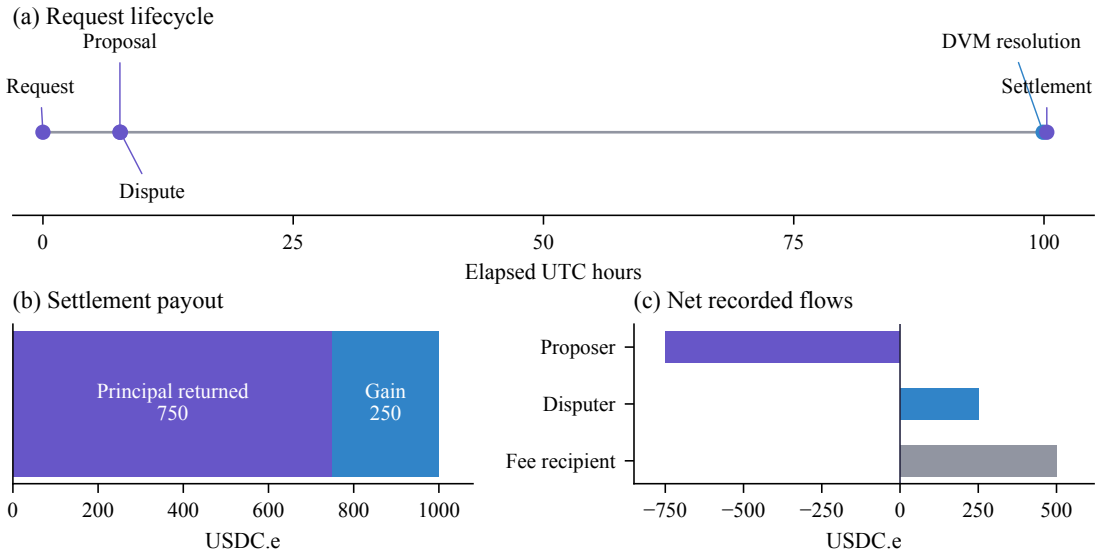


Figure 5. A linked UMA dispute. (a) Events on a proportional elapsed-time axis: Polygon events are purple and the Ethereum resolution is blue. The proposal and challenge are close in time. The interval between challenge and resolution includes unobserved intermediate transport steps. (b) The winner's gross payout separates principal from gain. (c) Participant and fee-recipient net flows balance in USDC.e, before additional costs.

7 Research use and interpretation

Research use should begin with a mechanism-specific question and an explicitly defined observation unit. Table 10 connects the eight branches to research questions, available observations and additional evidence requirements. These mappings identify opportunities for reuse rather than report tests of the proposed hypotheses. Analyses should retain implementation version, actor role, asset, economic stage and observation window, and establish comparability for the particular measure being studied. A shared schema facilitates this work but does not, by itself, identify causal effects or make different native activities interchangeable.

DuckDB, PyArrow and compatible tools can query Parquet tables through column selection and filtering. The repository notebook provides a smaller reconstruction example; the catalogue and dictionaries document table relationships and variables. Monetary values require asset and decimal fields. Event order follows native chronology, not row position. Addresses identify accounts, not their owners.

Economic aggregation must distinguish configured rewards, accruals, entitlements, payments and applied penalties. Returned principal differs from gain, and net profit requires costs. Check related components before summation; keep USDC.e and USDC separate unless a justified conversion rule applies. Missing, unavailable and inapplicable values are not zeros. Retain censored lifecycles as needed.

The UMA task supports modelling subject to retrospective selection and decision-time constraints. Causal analysis requires an identification strategy and further evidence.

Repositories provide data, code, documentation and an issue tracker. Reuse should record input versions and the computational environment, distinguish archived reconstruction from historical reacquisition, and preserve native meanings when extending analyses or reporting corrections.

8 Data and code availability

The data are available through the project's Hugging Face repository.[1] The release contains eight protocol products, the UMA prediction task and daily measurements, together with source registers, evidence ledgers and validation materials. Its catalogue records table identities, variable definitions and relationships. Earlier experimental outputs remain archived separately from these ten products.

The processing repository provides reconstruction scripts, figure sources and a tutorial with bounded

[1] Data and documentation: https://huggingface.co/datasets/Oracle4CEG/OracleEconLab-Oracle-Incentives-v1/tree/atlas-utc-20261029.

Table 10. Research opportunities supported by the eight branches. Rows identify potential uses and evidence needed for stronger claims; they are not findings or tests of mechanism-design guarantees.

Branch / literature	Question	Available observations	Further evidence needed
UMA [4, 22]	How do challenges and settlement timing vary with incentive commitments?	Bonds, configured rewards, liveness and request lifecycles.	Monitoring opportunities, costs, factual labels and an identification design for incentive effects.
Chainlink [8, 9, 23]	How do participation and reward realization relate to service provision?	Stake operations, reward claims and selected ETH/USD updates.	Node–service mappings, historical obligations and external references.
Tellor [4, 24]	How is accountability executed in observed disputes?	Reports, disputes, votes, payments and restrictions.	Broader comparable disputes, challenge opportunities and independent factual outcomes.
Flare FTSOv2 [4, 25]	How are reward entitlements formed and realized?	Provider conditions, entitlements, claims and epoch signing policies.	Eligibility–claim matching; historical runtime binding and entity-level ownership for control analysis; external prices for accuracy.
Pyth [4, 26]	How do participation and economic realization vary across OIS regimes?	Retained publisher–epoch states, staking instructions and transfers.	Effective rule-change times, comparable windows and treatment of concurrent changes.
Chronicle [2, 27]	Which reporting and challenge pathways are observable?	Registered reports, challenges and challenge rewards.	Historical signer sets, thresholds and factual references; full compensation remains unobserved.
RedStone Push [2, 28]	How do signed publications translate into Push updates?	Publication and update records for registered adapters and feeds.	Historical update settings and external references; AVS economics requires separate acquisition.
DIA [4, 29]	How much of a withdrawal is capital recovery versus reward?	Staking withdrawals and principal/reward components.	Deposit–withdrawal matching and costs for position returns; feeder lineage for price-production analysis.

examples for all eight branches.[2] The tutorial includes an executed local notebook. Input versions and execution records distinguish archived reconstruction from fresh public-interface acquisition; independent-reader and Google-hosted execution remain pending. Appendix H describes the reproducibility boundaries.

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A Scope and sources

A.1 Selection and scope boundaries

Branch selection combines a measurement question with identifiable historical evidence. UMA and Tellor cover contestable reporting and adjudication; Chainlink covers staking and selected feed service; Flare covers entitlements and claims; Pyth covers publisher–epoch incentives and staking. Chronicle, RedStone and DIA add narrower challenge, Push-update and withdrawal observations. Table 3 defines these boundaries. The registry is not exhaustive or probabilistic; exclusions apply only to this release.

Economic comparison requires an interpretable native unit, identifiable implementation and source. Monetary amounts also require an asset and accounting stage. Signed updates can describe delivery without measuring publisher payments; withdrawals can separate principal and reward without proving price accuracy. These limits govern the research opportunities in Table 10.

A.2 Historical implementations and source register

Table S1 records the principal contracts, programs and source versions. The accompanying source register describes their network roles, observation units, historical boundaries and inclusion criteria, including the Tellor modules and Chainlink feed proxy. Acquisition records preserve the connection between each sampled implementation and the evidence used to reconstruct it [30].

Deployment, scan and observation boundaries remain separate. Chronicle and RedStone searches begin at block 0, a search bound rather than a deployment block. Pyth's archive starts at a transaction timestamp rather than an EVM block. A pinned repository or cutoff runtime does not prove an unchanged historical implementation. Deployment times and implementation intervals require supporting metadata.

Table S1. Historical implementation and source register. Declared deployment blocks and retained source versions reproduce the deposited metadata; they are not newly verified deployment or runtime histories. Full addresses, source commits and unresolved fields are preserved in the accompanying files.

Branch	Principal identifier and chain role	Retained source, boundary and limitation
UMA	Polygon (137), OOv2: `0xee3afe347d5c7431 7041e2618c49534daf887c24`. Ethereum (1), VotingV2: `0x004395edb43efca9885cedad 51ec9faf93bd34ac`. Registered adapters and tunnels are listed separately in the contract CSV. Roles: verification, settlement and delivery, by implementation.	Retained UMA source `a16ee53125c4` and adapter source `8b76cc9e0d46`. Declared OOv2 deployment: Polygon block 29,786,052; earliest registered adapter: 23,569,780; cutoff: 89,439,336. DVM: 16,697,232–25,433,938. Historical runtime intervals are not fully established by cutoff code hashes.
Chainlink	Ethereum (1), community pool: `0xbc10f2e8 62ed4502144c7d632a3459f49dfcdb5e`; operator pool: `0xa1d76a7ca72128541e9fca cafbda3a92ef94fdc5`; ETH/USD proxy: `0x5f4ec3df9cbd43714fe2740f5e361615 5c5b8419`. Roles: staking accounting and feed delivery.	Staking v0.2 application configuration and ABI; retained phase-specific feed evidence. Pools start at declared deployment block 18,572,190; vault at 18,621,067; controller at 18,621,068; cutoff 25,433,938. Exact historical frontend/code version bindings remain incomplete.
Tellor	`tellor-1` application modules, ABCI `new_report` events and block results. Roles: reporting, adjudication and economic accounting on Tellor Layer.	Retained Layer source `943a2709ef0a`. Registered heights 1–19,890,860; first report at 317. Public opening is not genesis. The pre-opening network stage and completeness across historical module changes require separate evidence.

Continued on next page

Table S1 continued

Branch	Principal identifier and chain role	Retained source, boundary and limitation
Flare FTSOv2	Flare (14), RewardManager: `0xc8f55c5aa2c752ee285bd872855c749f4ee6239b`; VoterRegistry: `0x2580101692366e2f331e891180d9ffdf861fce83`. Roles: provider/registry evidence and reward accounting.	Reward artifacts: `fsp-rewards` commit `56dccda13597`. Declared deployment blocks 29,549,020 and 29,549,006, respectively; cutoff 64,054,789. Closed epochs 228–410; provider conditions start at 251. Aggregate claims need not identify every component.
Pyth	Solana mainnet OIS program: `pyti8TM4zRVBjmarcgAPmTNNAXYKJv7WVHrkrm6woLN`. Role: staking and transfer accounting; Pythnet-derived states are retained inputs, not a full independent chain scan.	Retained governance source `68a9a36ec3d4`. Transaction timestamps: 2024-09-11 19:53:53 to 2026-06-30 23:11:29 UTC. State epochs 2898–2946 cover 49 closed epochs. A later snapshot does not by itself establish historical decision-time availability.
Chronicle	Ethereum (1): 72 validated Scribe/ScribeOptimistic emitted-event contracts, individually identified in the source register. Roles: report validation, delivery and challenge-reward accounting.	Source `12ff06ca7881`; validator registry `f06e0a8209fc`. Registered scan: blocks 0–25,433,938. Zero is a scan start, not deployment. Historical deployment dates and per-interval implementation bindings are not established for every address.
RedStone Push	Ethereum (1): 9 Push adapters and 91 feed contracts in the frozen registry, with individual addresses recorded in the source register. Roles: signed-publication verification and delivery.	Source/manifest commit `95f8cb9fd00b`. Registered scan: blocks 0–25,433,938; deployment dates are separate. First retained update is not first protocol launch. Pull modes, other networks and full AVS economics are outside this branch.
DIA	Lasernet (1050), staking: `0x677Cf1299c367F6cf6F3E1669aCC18Fd059a5919`; wrapped-DIA address is retained in the contract CSV. Role: staking accounting and principal/reward payment.	Staking-application configuration and ABI retained in the manifest. Declared staking deployment: block 3,874,186; cutoff: 31,302,665. Deployment UTC time and a complete historical runtime-version mapping are not established by this summary.

A.3 Evidence for the ecosystem figure

Figure 1 uses first and last timestamps from the tables named in Table S2, derived from pinned coverage metadata [30]. These extents do not establish full branch histories or continuous coverage. Pyth uses epoch-end labels, while its transaction archive starts earlier. Tellor reports predate the public-opening announcement. Chainlink feed and DIA withdrawal histories need not start at staking deployment or launch.

Table S2. Selected-table timestamp extents used in Figure 1. Dates are UTC calendar dates, not deployment or scan bounds. Exact UTC timestamps and table identifiers are supplied with the figure source data.

Branch	Selected observation	First retained date	Last retained date
UMA	Request rounds	2022-11-04	2026-06-30
Chainlink	ETH/USD service windows	2023-11-14	2026-06-30
Tellor	Micro-reports	2025-05-09	2026-06-30
Flare FTSOv2	Reward-claim events	2024-10-03	2026-06-30
Pyth	Publisher epoch-end labels	2025-07-24	2026-06-25
Chronicle	Validator / challenge events	2023-08-28	2026-06-30

Continued on next page

Table S2 continued

Branch	Selected observation	First retained date	Last retained date
RedStone Push	Push update events	2023-12-07	2026-06-30
DIA	Staking withdrawals	2025-07-08	2026-06-18

Table S3 identifies dated historical markers at their source precision. Announcements, public access and deployment remain distinct: UMA's optimistic-oracle launch, for example, does not date OOv2 deployment. Each marker requires a source identifying the event and its date.

Table S3. Historical markers in Figure 1, dated at source precision. Announcements do not establish deployment. No verified RedStone first-Push milestone is available for inclusion.

Branch	Date / precision	Event and primary source
UMA	2018-12	Whitepaper; Project / network history [31].
UMA	2021-05-13	Optimistic Oracle; Service / public access [32].
Chainlink	2019-05-30	Ethereum mainnet; Service / public access [33].
Chainlink	2023-11-28	Staking v0.2; Mechanism announcement / access [34].
Tellor	2019-08-05	Ethereum mainnet; Service / public access [35].
Tellor	2025-08-04	Layer public access; Mechanism announcement / access [36].
Flare FTSOv2	2022-07-14	Flare genesis; Project / network history [37].
Flare FTSOv2	2024-09-16	FTSOv2 announcement; Mechanism announcement / access [38].
Pyth	2021-08-26	Solana feeds; Service / public access [39].
Pyth	2024-09-02	OIS introduction; Mechanism announcement / access [40].
Chronicle	2017	Maker oracle roots; Project / network history [41].
Chronicle	2023-09-05	Public access; Service / public access [41].
DIA	2025-03-26	Lasernet mainnet; Project / network history [43].
DIA	2025-06-24	Staking launch; Mechanism announcement / access [42].

The chain matrix distinguishes the sampled mechanisms from the wider services described in protocol documentation [22–29]. Its historical scope remains defined by the sampled implementations. Pythnet identifies retained inputs used in the Pyth reconstruction; it does not denote a complete chain scan. Unmarked cells indicate relationships outside the documented figure scope.

B Temporal harmonization

B.1 Native clocks and observation boundaries

Temporal coverage is defined for each sampled implementation and observation type. Its lower boundary may reflect contract deployment, a change in protocol rules, archive availability or the retention limits of a state interface. These boundaries are distinct from the first event found in a table. Table S4 summarizes the reasons for the observed scope; Appendix A provides the implementation and source details.

Table S4. Native clocks and lower boundaries. First observations do not establish deployment dates.

Branch	Native clock and observation	Starting boundary and limitation
UMA	Polygon event blocks; Ethereum DVM blocks. Request-reference timestamps remain separate.	Version-specific registered contracts and scans. The native DVM inputs lack explicit event clocks; only exact retained block matches are recovered.
Chainlink	Staking event blocks; ETH/USD `updated_at` payload and separate execution time.	Staking v0.2 deployment-bound collection; feed history is phase-specific. The native feed table and the common adjacent-window projection have different populations.
Tellor	Millisecond report time; aggregate index time; dispute and execution clocks, some with subsecond text.	tellor-1 scan through height 19,890,860. First retained report is at height 317; the later public-opening announcement does not date that event.
Flare FTSOv2	Reward-epoch support, claim execution time and cutoff state remain separate.	Retained reward epochs 228–410; conditions from 251. RewardManager and VoterRegistry have separate deployments. Epoch 411 straddles the cutoff and is outside the closed-epoch panel.
Pyth	Solana transaction time; publisher-epoch support; later state snapshot.	Transaction history predates the retained state panel. Its first epoch-end label is later than the beginning of its support interval. The circular buffer does not preserve full deployment history.
Chronicle	Ethereum execution block and separate oracle payload time.	Validated historical contract set, scanned from block 0 to the cutoff. The scan boundary and first retained event do not establish deployment.
RedStone Push	Ethereum Push execution block and separate payload time.	Pinned Push contracts, scanned from block 0. Pull and AVS activity are outside this observation window.
DIA	Lasernet staking event and verified withdrawal execution times.	Deployment-bound staking scan from block 3,874,186. The first withdrawal does not date deployment or the beginning of staking activity.

The interpretation of time also depends on the native object. A UMA request-reference timestamp contributes to request identity, whereas the block timestamp dates the recorded event. Tellor report, aggregate and dispute clocks refer to different stages of reporting and adjudication. An epoch or state snapshot represents temporal support or a state observation rather than an instantaneous event.

B.2 UTC representation and epoch support

UTC conversion preserves native values, units and precision. Table S5 illustrates second and millisecond clocks, epoch intervals and missing-time handling. Native identifiers and execution order are retained; nanosecond storage does not increase source precision.

Missing EVM times are recovered from consistent same-chain, same-block timestamps, with provenance distinguishing recovered and source-provided clocks. Conflicting or unmatched blocks remain undated.

Epoch-based observations are represented by their native start and end boundaries. The retained Flare and Pyth reward intervals have different durations, and neither interval determines when a payment occurred.

Table S5. Native-to-UTC examples from retained observations. Epochs use $[a,b)$ intervals; missing values remain explicit. Record identifiers and source matches accompany the examples.

Observation	Native value	UTC representation	Interpretation
E1 UMA request	1667593230 s	2022-11-04 20:20:30	Exact second conversion; request-event block time.
E2 Tellor report	1746801845665 ms	2025-05-09 14:44:05.665	Exact millisecond conversion; fractional seconds retained.
E3 Pyth epoch	[1752710400, 1753315200) s	Start: 2025-07-17 End: 2025-07-24 Both at 00:00:00	Epoch support interval; no event or payment time inferred.
E4 Chainlink staking	Missing; Ethereum block 18,671,606	2023-11-28 17:27:47	Recovered from a retained Chronicle record for the same Ethereum block.
E5 Chainlink staking	Missing; Ethereum block 18,621,068	Missing	No retained block-time match; value remains missing.

Figure S1 illustrates their alignment near the observation cutoff. A timestamp on a shared epoch boundary belongs to the later interval under the half-open convention. Later state snapshots preserve their own acquisition context and do not establish when historical values became available to participants.

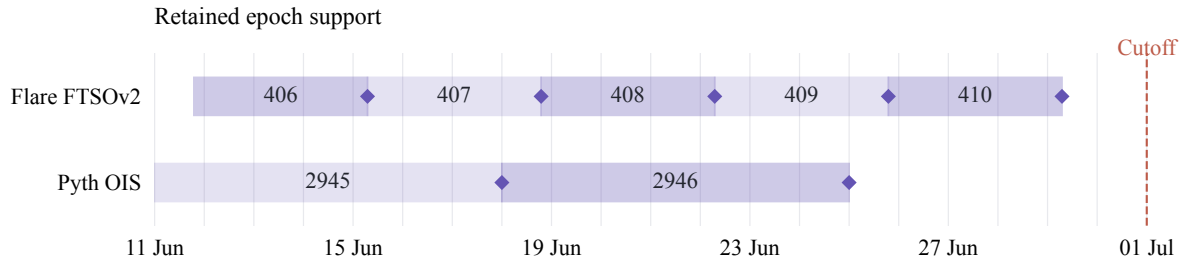


Figure S1. Native epochs near the cutoff. Flare and Pyth bars span three and a half and seven days, respectively; diamonds mark epoch-end labels against a daily grid. Labels denote observation intervals, not payments.

B.3 Temporal availability and coverage

Timestamp availability is the proportion of retained observations with a valid clock. Historical coverage additionally requires evidence about the period in which a mechanism was active and the intervals successfully collected. A sequence with timestamps on every row can therefore still contain acquisition gaps. Coverage remains unspecified when the underlying observation period cannot be established.

Table S6. Timestamp availability within retained populations after exact block matching. The Chainlink subset covers staking, withdrawals, reward payments, reward finalization and forfeiture distributions.

Observation population	Available	Remaining limitation
UMA DVM voter accruals	10.9%	Execution time is unavailable for unmatched blocks.
Chainlink native staking logs	6.0%	Retained block references cover only part of the log history.
Chainlink analytical subset	5.0%	Temporal analysis remains restricted to the dated observations.

Native and UTC timestamps and record selections agree within the tested windows. Subsecond values and native order are preserved, and epoch duration and membership checks support the interval representation.

These tests assess processing consistency without establishing clock accuracy or continuous acquisition.

Temporal annotations and the original common representation are separately versioned. Analyses should use a consistent representation and restrict comparisons to compatible observation periods. Daily economic measures further require defined participant roles, assets, economic stages and exposure. Epoch totals should remain attached to their support intervals unless the data justify a more detailed temporal allocation.

C Product organization and metadata

The ten logical products group observations by mechanism or reuse task. They do not denote ten physical files or ten independent samples. Native and harmonized representations may describe the same event, and the two derived products draw on the protocol branches. The catalogue therefore records the product membership of each table together with its row unit, source relationship and version.

Table S7. Ten logical data products and their principal observation units.

Product	Principal units	Main relationships
UMA	Request, vote, voter accrual and stake event	Polygon request–Ethereum adjudication; settlement–transfer
Chainlink	Feed update, service interval and staking event	Staking or reward event–LINK movement
Tellor	Report, aggregate, dispute and payment	Report–dispute; tip funding–withdrawal
Flare FTSOv2	Provider–epoch, entitlement and claim	Epoch policy–registration; entitlement–claim evidence
Pyth	Instruction, reward transfer and publisher–epoch	OIS instruction–custody transfer; historical epoch state
Chronicle	Update, challenge and challenge reward	Registered implementation–event; challenge–payment
RedStone Push	Adapter/feed update	Registered delivery contract–update
DIA	Stake action and withdrawal position	Position–principal and reward components
UMA prediction	One eligible request	Proposal-time feature–availability record–resolution label
Daily measurements	Date–branch–role–asset–stage–measure	Native eligible population–daily aggregate; epoch companions separate

Field dictionaries specify types, units, economic stage, source keys, missingness and calculation rules. Raw monetary amounts remain integer-valued strings, while UTC clocks distinguish event time from reference time, deadlines, epochs and snapshots. Croissant recordsets describe physical tables and link them to their logical product; the product count is consequently different from the recordset count. The dataset card, table catalogue and tutorial use the same product identifiers.

D Daily measurement definitions

D.1 Feasibility and coverage

Feasibility is assessed for each branch and measurement family before aggregation. A derivable quantity is supported within a stated retained population; this designation does not imply continuous acquisition or applicability to every service offered by the protocol. A quantity requiring additional acquisition has a plausible source or linkage requirement. Inapplicable quantities do not describe the sampled unit, whereas unknown quantities lack sufficient evidence to resolve their status. Figure S2 summarizes selected families; the accompanying matrix includes the full definitions and qualifications.

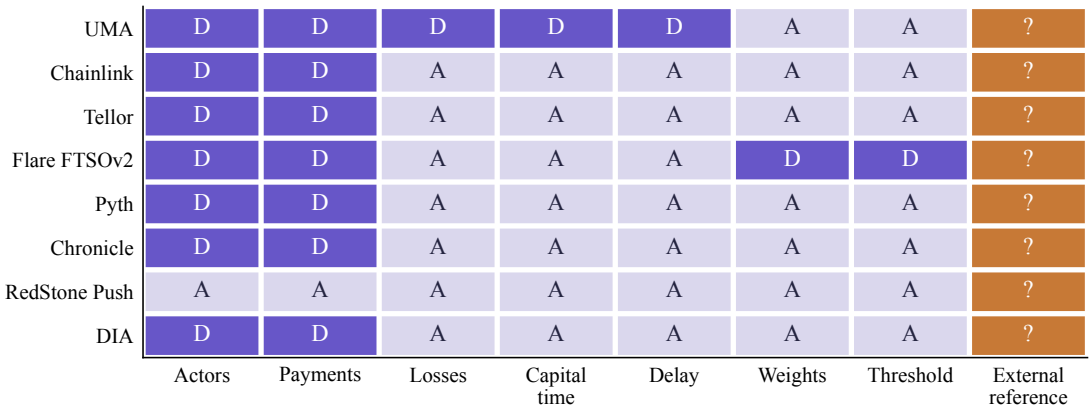


Figure S2. Measurement feasibility by branch. D: derivable in the retained population; A: additional acquisition or linkage needed; dash: inapplicable; question mark: unresolved. Flare weights and thresholds retain epoch support. RedStone Push covers delivery contracts; publisher participation and AVS economics require separate evidence.

D.2 Aggregation and denominators

For an eligible event population $\mathscr{E}_g$ with grouping key g, a daily amount is

$$A_{dg} = \sum_{i \in \mathscr{E}_g : t_i \in [d, d+1)} a_i. \tag{4}$$

The grouping key retains branch, implementation, actor role, asset and stage. Integer amounts are summed before decimal conversion. Eligible retained events define the denominator without establishing complete acquisition. Untimed observations cannot enter daily groups and remain in the missingness accounting.

Participation counts distinct observed addresses within a specified role. Delivery-contract activity in RedStone is recorded as contract activity, and Flare reward owners are distinguished from registered providers. Reward inequality uses positive observed recipient totals within the same day, asset and stage. The top-decile share includes the largest $\lceil 0.1n \rceil$ recipient totals. A recipient with no recorded payment is outside that recipient population; its absence is not a verified zero reward.

UMA completion delay is measured from the request event to the settlement event and grouped by settlement day. Its denominator consists of completed, flow-reconciled requests. A completion fraction would additionally require a defined obligation and a comparable risk set, so it is not inferred by dividing daily settlements by daily requests. Escrow exposure is computed from the running balance implied by the request's token movements. A request is included only when its lifecycle transactions map unambiguously to that request, all included transfers are timed, balances remain nonnegative and the closing balance is zero. With balance b_{ik} held between successive transfer times t_{ik} and $t_{i,k+1}$, daily exposure is

$$K_{di} = \sum_k b_{ik} \left| [t_{ik}, t_{i,k+1}) \cap [d, d+1) \right|. \tag{5}$$

The stored quantity uses raw token-seconds; conversion to token-hours divides by the asset precision and 3,600. Fees cease contributing when they leave escrow, while funded rewards contribute for as long as they remain held. The resulting measure describes observed contract balances for the eligible closed-request cohort. It does not establish participant wealth, total protocol collateral or the amount subject to slashing.

D.3 Historical signing policies

Flare epoch artifacts record signers, integer weights, a threshold and the starting voting round, matched to retained voter registrations. Normalized weights w_i give concentration $H = \sum_i w_i^2$ and effective address count $1/H$. The table reports the fewest largest-weight addresses needed to reach or exceed the threshold under weak and strict conventions. An operative security threshold requires historical contract and runtime evidence. These address-level measures do not identify owners, independent organizations or attack feasibility.

Entitlements, signing weights and deposited capital retain different units and roles. Epoch values keep their support intervals; uniform daily allocation does not turn reward factors into payments. Signatures and aggregate agreement do not independently measure price accuracy.

D.4 Temporal sensitivity and lifecycle follow-up

Weekly aggregation preserves daily event counts and integer amounts within each typed group. A separate comparison restricts observations to the intersection of the branches' retained calendar envelopes. This restriction aligns the outer dates but cannot establish continuous coverage within them. Estimates requiring verified complete intervals therefore remain unavailable. For epoch entitlements, an allocation comparison contrasts the epoch-end label with proportional overlap across UTC days. These are alternative representations of the same entitlement; neither is included among observed daily payments.

UMA follow-up retains settled and unresolved requests until settlement or the common cutoff. Figure S3 shows completed-request delays and unresolved shares by request year. Delays are conditional on settlement, and cohorts have unequal follow-up. Neither panel estimates protocol-wide timely-resolution probabilities.

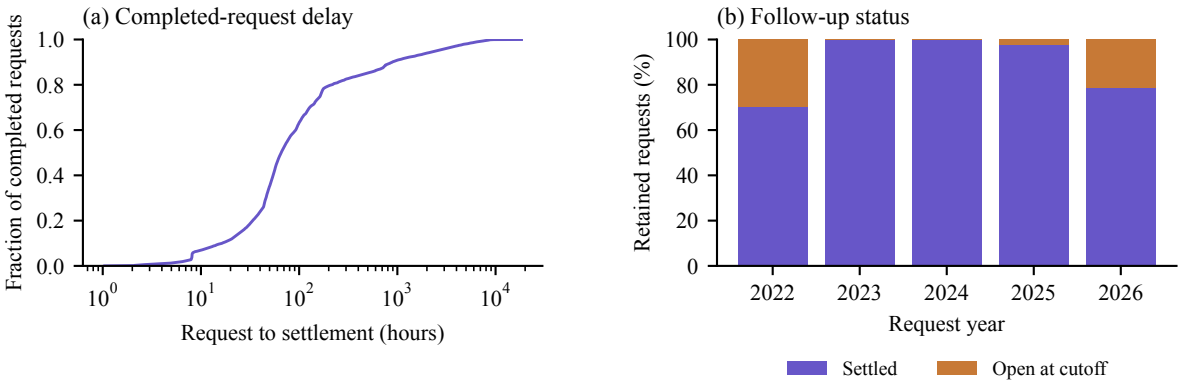


Figure S3. UMA request follow-up. (a) Empirical cumulative distribution of elapsed request-event to settlement time among completed requests. (b) Settlement status at the common cutoff, grouped by the year of the request event. Requests still open at the cutoff are right-censored; an unobserved later settlement is not classified as failure.

E Semantic rules and assistance disclosure

The correction ledger links changed fields to native objects, historical rules and prior interpretations. UMA challenge status requires a dispute; settlement requires completion and the relevant token flow. Winner payouts separate principal from the unburned losing bond. Tellor SUPPORT accepts the challenge; an AGAINST outcome does not imply reporter loss merely because stake previously entered escrow.

VotingV2 requires a further distinction between request-level accrual and application to stake. A signed accrual records an adjustment associated with a voting request. The later application event concerns the net change actually applied to a stake position. The ledger retains the accrued amount and leaves a request-level applied loss unknown when application evidence is not matched. Adding the accrual to a subsequent net application would count stages of the same accounting process twice.

Historical binding is assessed separately from arithmetic. Evidence is stronger for VotingV2 than for OOv2 and Tellor. A commit identifies inspected code but does not prove which binary executed at each block. Table 9 reports corrections only for the tested populations.

Codex assisted with source inspection, processing code, manuscript editing and figures; the assistance record documents its contributions and available model information. Automated checks assess consistency and reconstruction, while economic interpretation remains limited by the historical source evidence.

F UMA task definition and feature dictionary

The task contains 810 retrospectively eligible requests, of which 466 resolve to a value different from the proposal and 344 retain the proposed value. A changed value is an institutional adjudication label. The

cohort is restricted to challenged, linked and reconciled requests, so these proportions do not estimate the prevalence of incorrect proposals in the full request population.

Excluding early expiration leaves 413 requests with 69 changed proposals; restricting resolutions to binary zero or one leaves 402 requests with 59 changes. These cohorts support sensitivity analysis. The main task keeps native classes because early expiration differs from either binary outcome.

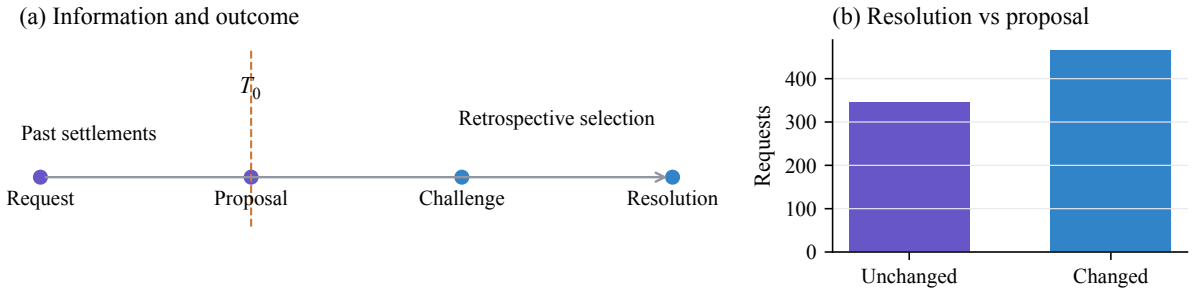


Figure S4. Information boundary and target of the UMA task. (a) T_0 is the proposal event; prior-history features use earlier settled requests. Challenge and resolution determine retrospective eligibility and the outcome. (b) Counts compare the native proposed and resolved integers, retaining special resolution classes.

Let B, F and R denote the effective bond, final fee and configured reward in the asset's smallest units. For address a in a specified role, N_a counts retained prior settlements, D_a counts those challenged, and J_a counts those whose challenged proposal was rejected. Prior settlements must precede the proposal in block, transaction and log order. Proposer and requester histories are separate and bounded by the retained records.

Table S8. Sixteen candidate features. Rates with $N_a = 0$ and ratios with $B = 0$ are missing. Parameter-derived entries require historical binding before they can be certified as available at the proposal.

Feature	Definition or unit	Evidence / missingness
Bond magnitude	$\log(1+B)$	Reconstructed effective bond; historical parameter binding.
Final-fee magnitude	$\log(1+F)$	Native fee; missing if absent.
Reward magnitude	$\log(1+R)$	Configured adapter reward; historical binding.
Reward–bond ratio	R/B	Same-asset amounts; undefined for zero bond.
Liveness	Expiry minus proposal, hours	Proposal event and recorded expiry.
Proposal latency	Proposal minus request event, hours	Execution clocks, not request-reference time.
Ancillary length	Decoded byte count	Native request payload.
Proposer prior count	N_a	Earlier settlements in proposer role.
Proposer dispute rate	D_a/N_a	Missing when no prior settlement is retained.
Proposer rejection rate	J_a/N_a	Rejected challenged proposals among prior settlements.
Requester prior count	N_a	Earlier settlements in requester role.
Requester dispute rate	D_a/N_a	Missing when no prior settlement is retained.
Requester rejection rate	J_a/N_a	Same-role prior settlement outcomes.
Registered adapter	Binary membership	Requester contract registry.
Proposed price class	Zero, one, half, early expiry, other	Native proposed integer; no forced binary recoding.
Adapter version	Registered implementation class	Historical source binding required.

Feature availability is recorded for each request and feature. Native event fields, reconstructed past histories and parameters requiring historical binding have distinct statuses. Fields used only to establish the eventual label remain outside the feature allowlist. Missing rates are retained as missing values so that subsequent users can choose and document their own preprocessing. Figure S5 presents the feature distributions; no predictive model is fitted as part of this data description.

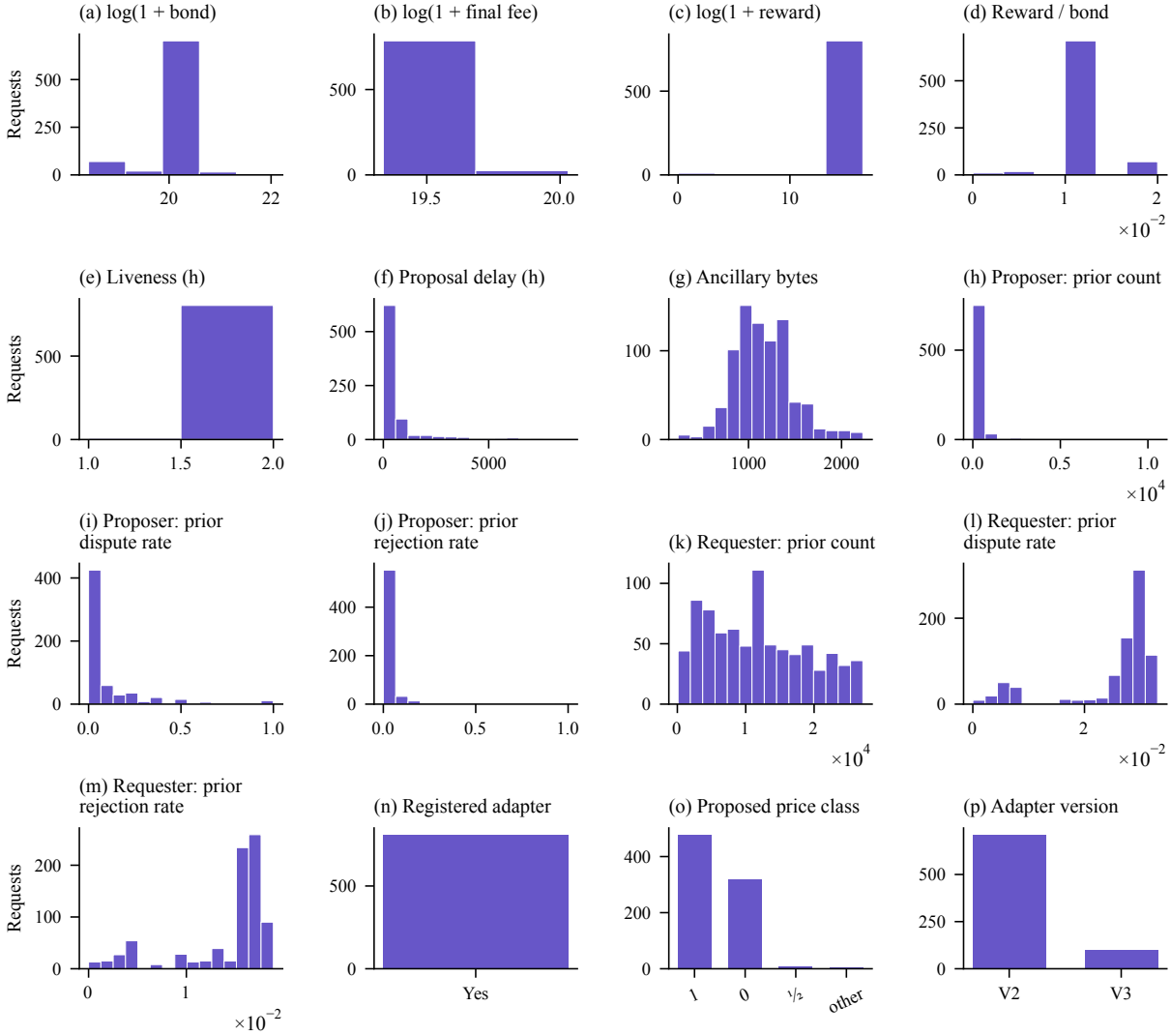


Figure S5. Distributions of the sixteen task features. Histograms use nonmissing observations on the units in Table S8; categorical panels retain the recorded classes. Prior-history rates exclude requests with no retained earlier settlement in that role. Detailed missingness and availability statuses accompany the task data.

G Cross-chain event register

The worked case is a single OOv2 request linked to one VotingV2 request. Table S9 locates the five observed endpoints. Full contract addresses, request identifiers, transaction hashes and receipt evidence accompany the register. A log position is interpreted within its transaction and chain; equal-looking identifiers on different chains are not joined without the registered ancillary-data transformation.

The Polygon request, proposal and dispute are decoded from OOv2 events; Ethereum's resolution is decoded from VotingV2. The Polygon settlement retains the same resolved value. Native request hashing and ancillary-data transformations support the identity match. The five retained transactions do not provide a complete sequence of relay acceptance, processing and return operations, so the interval between dispute and resolution is not attributed solely to voting or solely to transport.

Accounting separates both deposits, the winner's gross payout and the protocol fee. Principal return and

Table S9. Observed endpoints of the worked UMA dispute. UTC coordinates identify actual case events. Explorer links locate the transactions; retained receipts provide the reproducible evidence.

Event	Chain	UTC	Block / log	Evidence
T1 Request	Polygon	2024-10-29 20:05:50	63645613 / 196	Transaction
T2 Proposal	Polygon	2024-10-30 03:47:16	63658346 / 327	Transaction
T3 Dispute	Polygon	2024-10-30 03:49:54	63658420 / 397	Transaction
T4 DVM resolution	Ethereum	2024-11-03 00:00:35	21103395 / 141	Transaction
T5 Settlement	Polygon	2024-11-03 00:22:35	63813887 / 104	Transaction

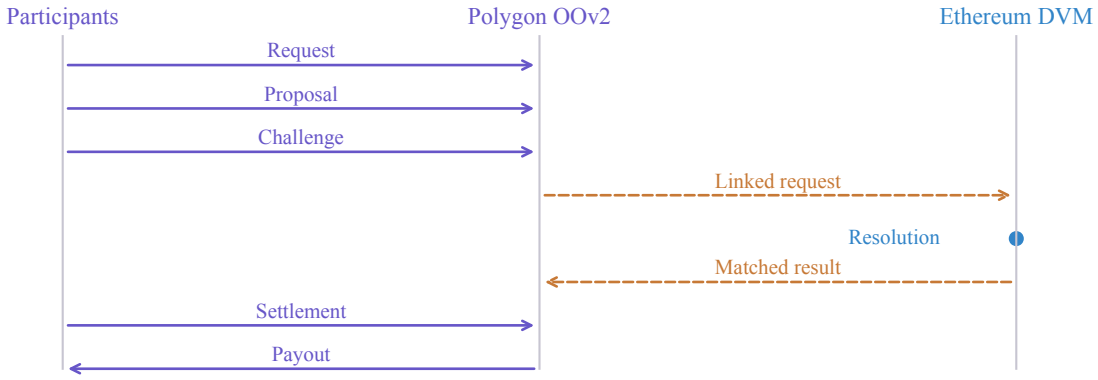


Figure S6. Evidence links in the worked case. Solid arrows denote observed participant actions or transfers. Dashed arrows connect matched request identities and outcomes across chains; they do not represent an independently reconstructed relay path. Vertical order is schematic. Elapsed times are shown in Figure 5.

gain decompose one payout and must not be added again as transfers. The configured reward is tracked separately. Early expiration retains its native meaning; the proposition lacks independent factual adjudication.

H Reconstruction and versioned evidence

Reproducibility is organized around the relationship between retained inputs, deterministic transformations and named outputs. Input fingerprints bind a calculation to the files it used. Source registries identify the interfaces and historical boundaries, while result manifests connect tables and figures to the relevant processing rules. A successful replay establishes consistency for those inputs and transformations.

The tutorial provides one bounded example per branch, checking retained inputs, native units and clocks. Fresh historical queries depend on archive retention and provider behaviour. Retrieving an individual receipt or block does not establish continuous acquisition across the registered interval.

The reproduction record documents commands, environment versions, validation and acquisition limits. Earlier model experiments remain separate from the revised feature definitions. The catalogue and checksums identify this revision's materials and distinguish them from previous public deposits.

I Measurement glossary

Table S10. Terms used to distinguish protocol activity, economic states and evidence.

Term	Meaning
Branch	Registered mechanism and implementation scope, with specified chain roles.
Observation unit	Native object counted or measured, such as a request, claim or provider–epoch.
Configured incentive	A rule or parameter describing a potential reward or penalty.
Accrual	An accounting adjustment associated with an activity, before separate application or payment evidence.
Entitlement	A recorded claim to an amount under the applicable eligibility rules.
Realized payment	A qualifying transfer or execution record under a stated branch-specific rule.
Applied loss	A supported reduction or forfeiture; not inferred from every negative accrual.
Principal return	Recovery of posted capital, distinct from income.
Escrow exposure	Observed token balance integrated over time, including funded rewards while held and retaining the eligible request population.
Policy weight	A mechanism-specific influence measure; not automatically collateral or organizational independence.
Coverage state	Evidence about what was observable and collected for a particular population or interval.
Protocol adjudication	An institutional resolution; external truth requires independent evidence.